

一. 简答题

1. Ans.

区别: 信息是一个抽象概念, 可以脱离信息系统的各个组成和阶段独立存在;

数据是一个具象概念, 往往是信息的具体表现形式, 并可能随着载荷它的物理设备的形式而改变。

联系: 信息与数据不可分离, 信息由物理介质承载的数据表达, 数据中所包含的含义就是信息。

2. Ans.

因为异或问题的数据是线性不可分的。

可以采用核函数方法, 把低维数据映射到高维, 使之线性可分。

3. Ans.

长短期记忆网络中, 门控机制有输入门, 输出门和遗忘门, 以及隐藏的细胞状态。其中细胞状态存储着先前的信息, 而遗忘门控制着获取先前信息的比例。

与认知心理学的关联:

在人的记忆中, 就有长时记忆和短时记忆这两部分。

4. Ans.

区别: 欠拟合是由于模型相对于数据过于简单, 特征维数不够多, 而导致训练集与测试集的 error 都较大。

过拟合是由于模型相对于数据过于复杂, 拟合函数考虑了噪声等不必要的数据关联, 而导致训练集 error 小, 测试集误差大。

避免过拟合: ① 引入验证集, 做 early-stopping.

② 在 linear 层做 dropout.

③ 扩增数据集.

5. Ans.

混淆矩阵:

实际 \ 预测	正	反
正	7	3
反	2	4

$$\text{Recall} = \frac{7}{7+3} = 70\%$$

$$\text{Precision} = \frac{7}{7+2} = 77.8\%$$

$$\text{TPR} = \frac{7}{7+3} = 70\%$$

$$\text{FPR} = \frac{2}{2+4} = 33.3\%$$

$$\text{Accuracy} = \frac{11}{16} = 68.8\%$$

6. Ans.

① Sigmoid 激活函数, $f(x) = \frac{1}{1+e^{-x}}$, 前向传播运算量较大,

较易发生梯度消失, 不过较易求导. 值域: $(0, 1)$

② Tanh 激活函数: 性质与 Sigmoid 类似, 但解决了 Sigmoid 输出零点不对称的问题.

值域: $(-1, 1)$

③ ReLU 激活函数: 计算简单, 收敛速度快, 缓解过拟合问题.

但可能会使梯度过大, 而导致神经元死亡. 值域: $[0, +\infty)$

∵ 输出为负 ∴ 可能是 Tanh.

7. Ans.

三种感受野 | 同心圆感受野
| 简单感受野
| 复杂感受野

三者关系：排成一条线的同心圆对一定朝向的条状物敏感。

若干同朝向的简单感受野可聚合为复杂感受野。

举例：Gabor 滤波器。

8. Ans.

$H_{in} = W_{in} = 227$, $C_{in} = 3$, $Co_{out} = 96$, $K_H = K_W = 7$, $s = 2$, $p = 1$, 则

$H_{out} = W_{out} = \lfloor (W_{in} - K_W + 2p) / s \rfloor + 1 = 112 \Rightarrow$ 输出尺寸: $112 \times 112 \times 96$

参数量 = $Co_{out} \times (C_{in} \times K_H \times K_W + 1) = 14208$

2. 简单计算题

1. Ans.

定义 A := 该人确诊, B := 该人检测阳性, 则

$$\begin{aligned} P(A|B) &= \frac{P(B|A) P(A)}{P(B|A) P(A) + P(B|A^c) P(A^c)} \\ &= \frac{0.01 \times 0.8}{0.01 \times 0.8 + 0.99 \times 0.096} = 7.8\% \end{aligned}$$

2. Ans.

显然, 为 $L = -\log 0.8 = 0.223$

3. Ans.

$$\text{似然函数 } l(\mu, \Sigma; X) = \frac{1}{(2\pi)^{\frac{3}{2}} |\Sigma|^{\frac{1}{2}}} \exp[-(X-\mu)^T \Sigma^{-1} (X-\mu)]$$

$$\text{需求 } (\mu, \Sigma) = \arg \max l(\mu, \Sigma; X)$$

$$\text{化简得 } \begin{cases} \mu = \frac{1}{3}(x_1 + x_2 + x_3) \\ \Sigma = \frac{1}{3} \sum_{i=1}^3 (x_i - \mu)(x_i - \mu)^T \end{cases}$$

$$\Rightarrow \begin{cases} \mu = [-2, 2]^T \\ \Sigma = \frac{1}{3} \begin{bmatrix} 2 & -4 \\ -4 & 14 \end{bmatrix} \end{cases}$$

4. Ans.

$\because \lambda_2 = 15.82 > \lambda_1 = 2.58 \quad \therefore \begin{bmatrix} 0.3 \\ -0.4 \end{bmatrix}$ 为其主成分, 15.82 为该主成分对应的特征值

归一化主成分: $V_1 = \begin{bmatrix} 0.6 \\ -0.8 \end{bmatrix}$

$$\text{则 } Z_m = \langle (X-\mu), V_1 \rangle = 1.2$$

三. 计算题

1. Ans.

由 Pf. 不失一般性, 任取第 i 类与第 j 类 ($1 \leq i, j \leq 3, i \neq j$),

$$\text{有 } p(\omega_i|x) = \frac{p(\omega_i)p(x|\omega_i)}{p(x)}, \quad p(\omega_j|x) = \frac{p(\omega_j)p(x|\omega_j)}{p(x)}$$

$$\text{分类界面: } \log p(\omega_i|x) = \log p(\omega_j|x)$$

$$\Rightarrow \log p(x|\omega_i) = \log p(x|\omega_j)$$

类别先验相等

而 X 样本的似然函数满足正态分布且 Σ 一样.

$$\therefore p(x|\omega_i) = \frac{1}{(2\pi)^{\frac{1}{2}} |\Sigma|^{\frac{1}{2}}} \exp(-(x-\mu_i)^T \Sigma^{-1} (x-\mu_i))$$

$$p(x|\omega_j) = \frac{1}{(2\pi)^2 |\Sigma|^{1/2}} \exp(-(x-\mu_j)^T \Sigma^{-1} (x-\mu_j))$$

代λ得 $(x-\mu_i)^T \Sigma^{-1} (x-\mu_i) = (x-\mu_j)^T \Sigma^{-1} (x-\mu_j)$

化简: $\underbrace{-\mu_i^T \Sigma^{-1} x - x^T \Sigma^{-1} \mu_i + \mu_i^T \Sigma^{-1} \mu_i}_{= -2\mu_i^T \Sigma^{-1} x} = \underbrace{-\mu_j^T \Sigma^{-1} x - x^T \Sigma^{-1} \mu_j + \mu_j^T \Sigma^{-1} \mu_j}_{= -2\mu_j^T \Sigma^{-1} x}$

$$\Rightarrow 0 = (\mu_i - \mu_j)^T \Sigma^{-1} x - \frac{1}{2} (\mu_i^T \Sigma^{-1} \mu_i - \mu_j^T \Sigma^{-1} \mu_j) \quad \text{Done!}$$

(2) 代λ即得:

$$\Sigma^{-1} = \begin{bmatrix} 0.9 & -0.2 \\ -0.2 & 0.6 \end{bmatrix}$$

$$\mu_1^T \Sigma^{-1} \mu_1 = 0.6, \quad \mu_2^T \Sigma^{-1} \mu_2 = 0.9, \quad \mu_3^T \Sigma^{-1} \mu_3 = 1.9$$

$$\Rightarrow g_{1,2}(x) = 0.7x_1 + 0.4x_2 + 0.15$$

$$g_{1,3}(x) = -0.3x_1 + 0.4x_2 + 0.65$$

$$g_{2,3}(x) = -2x_1 + x_2 + 0.5$$

(3) 对 $[x_1, x_2]^T = [0.2, 1]^T$, 有

$$g_{1,2}(x) = 0.14 + 0.4 + 0.15 > 0 \Rightarrow \text{不在2类}$$

$$g_{1,3}(x) = -0.26 + 0.4 + 0.65 > 0 \Rightarrow \text{不在3类}$$

$\Rightarrow$ 其必属于1类.

2. Ans.

4. 使用前向递推法:

$$t=1: \alpha_1(S) = \pi_S B_S(D) = 0.3 \times 2 = 0.6$$

$$\left| \alpha_1(C) = \pi_C B_C(D) = 0.5 \times 0.7 = 0.35 \right.$$

$$\alpha_1(R) = \pi_R B_R(D) = 0.2 \times 0.4 = 0.08$$

$$t=2: \alpha_2(S) = \left[\sum_{i \in S} \alpha_1(i) A_{is} \right] b_S(W) = 0$$

$$\left| \alpha_2(C) = \left[\sum_{i \in S} \alpha_1(i) A_{ic} \right] b_C(W) = 0.0648 \right.$$

$$\alpha_2(R) = \left[\sum_{i \in S} \alpha_1(i) A_{ir} \right] b_R(W) = 0.098$$

$$t=3: \alpha_3(S) = [\sum_{i \in S'} S_2(i) A_{is}] b_s(D) = 0.0414$$

$$\left| \alpha_3(C) = [\sum_{i \in S'} S_2(i) A_{ic}] b_c(D) = 0.0335 \right.$$

$$\alpha_3(R) = [\sum_{i \in S'} S_2(i) A_{ir}] b_r(D) = 0.0341$$

$$\Rightarrow P_3(DWD) = \alpha_3(S) + \alpha_3(C) + \alpha_3(R) = 0.109$$

(2) 应用 Viterbi 算法:

$$t=1: \delta_1(S) = \pi_S B_S(D) = 0.3 \quad \varphi_1(S) = 0$$

$$\left| \delta_1(C) = \pi_C B_C(D) = 0.25 \quad \varphi_1(C) = 0 \right.$$

$$\delta_1(R) = \pi_R B_R(D) = 0.08 \quad \varphi_1(R) = 0$$

$$t=2: \delta_2(S) = \max_{i \in S'} [\delta_1(i) A_{is}] B_S(W) = 0$$

$$\left| \delta_2(C) = \max_{i \in S'} [\delta_1(i) A_{ic}] B_C(W) = 0.042 \right.$$

$$\delta_2(R) = \max_{i \in S'} [\delta_1(i) A_{ir}] B_R(W) = 0.063$$

$$\left| \varphi_2(S) = \operatorname{argmax}_{i \in S'} [\delta_1(i) A_{is}] = S \right.$$

$$\varphi_2(C) = \operatorname{argmax}_{i \in S'} [\delta_1(i) A_{ic}] = C$$

$$\varphi_2(R) = \operatorname{argmax}_{i \in S'} [\delta_1(i) A_{ir}] = C$$

$$t=3: \delta_3(S) = \max_{i \in S'} [\delta_2(i) A_{is}] B_S(D) = 0.0126$$

$$\left| \delta_3(C) = \max_{i \in S'} [\delta_2(i) A_{ic}] B_C(D) = 0.0118 \right.$$

$$\delta_3(R) = \max_{i \in S'} [\delta_2(i) A_{ir}] B_R(D) = 0.0151$$

$$\left| \varphi_3(S) = \operatorname{argmax}_{i \in S'} [\delta_2(i) A_{is}] = C/R \right.$$

$$\varphi_3(C) = \operatorname{argmax}_{i \in S'} [\delta_2(i) A_{ic}] = C$$

$$\varphi_3(R) = \operatorname{argmax}_{i \in S'} [\delta_2(i) A_{ir}] = R$$

$$\Rightarrow Q_3 = R \Rightarrow Q_2 = R \Rightarrow Q_1 = C$$

$$\Rightarrow \alpha_{1,2,3} = \{CAB\}$$

3. Ans.

4) 令 $f(p) = y \log p + (1-y) \log (1-p)$

则 $\frac{\partial f}{\partial p} = \frac{y}{p} - \frac{1-y}{1-p}$

而 $p = [1 + e^{-z}]^{-1} \Rightarrow \frac{\partial p}{\partial z} = p(1-p)$

$\Rightarrow \frac{\partial f}{\partial z} = \frac{\partial f}{\partial p} \frac{\partial p}{\partial z} = y(1-p) - (1-y)p = y-p$

而 $p = w_0 + w_1 x_1 + w_2 x_2$

$\Rightarrow \frac{\partial f}{\partial w_0} = \frac{\partial f}{\partial z} = y-p$

$\left| \frac{\partial f}{\partial w_1} = x_1 \frac{\partial f}{\partial z} = x_1(y-p) \right.$

$\frac{\partial f}{\partial w_2} = x_2 \frac{\partial f}{\partial z} = x_2(y-p)$

$\Rightarrow \frac{\partial L}{\partial w_0} = \sum_{i=1}^n (p(z_i) - y_i)$

$\left| \frac{\partial L}{\partial w_1} = \sum_{i=1}^n x_{1,i} (p(z_i) - y_i) \right.$

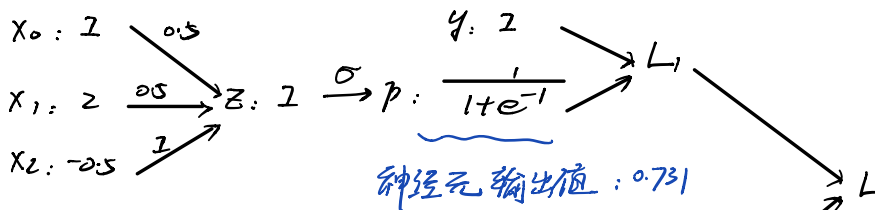
$\frac{\partial L}{\partial w_2} = \sum_{i=1}^n x_{2,i} (p(z_i) - y_i)$

$w_0 \leftarrow w_0 - \frac{\partial L}{\partial w_0} \lambda$

$w_1 \leftarrow w_1 - \frac{\partial L}{\partial w_1} \lambda$

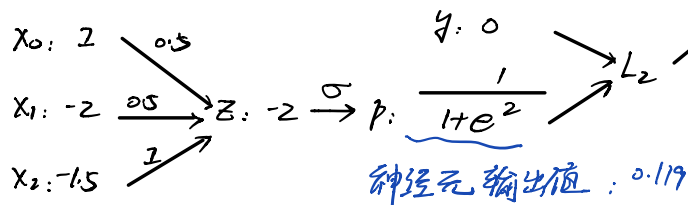
$w_2 \leftarrow w_2 - \frac{\partial L}{\partial w_2} \lambda$

2) 样本1:



神经元输出值: 0.731

样本2:



神经元输出值: 0.119

3) $\frac{\partial L}{\partial w_0} = 0.731 + 0.119 - 1 = -0.150$

$\frac{\partial L}{\partial w_1} = 2 \times (0.731 - 1) + (-2) \times (0.119 - 0) = -0.776$

$$\frac{\partial L}{\partial W_2} = -0.5 \times (0.731 - 1) + (-1.5) \times (0.119 - 0) = -0.044$$

$$\Rightarrow W_0 = 0.5 - (-0.150) \times 0.1 = 0.515$$

$$W_1 = 0.5 - (-0.776) \times 0.1 = 0.578$$

$$W_2 = 2 - (-0.044) \times 0.1 = 1.004$$

4. Aus.

$$D_1 = (1, 1)^T \Rightarrow z = (2, 0) \quad \text{类别 } 1$$

$$D_2 = (2, 1)^T \Rightarrow z = (5, 1) \quad 1$$

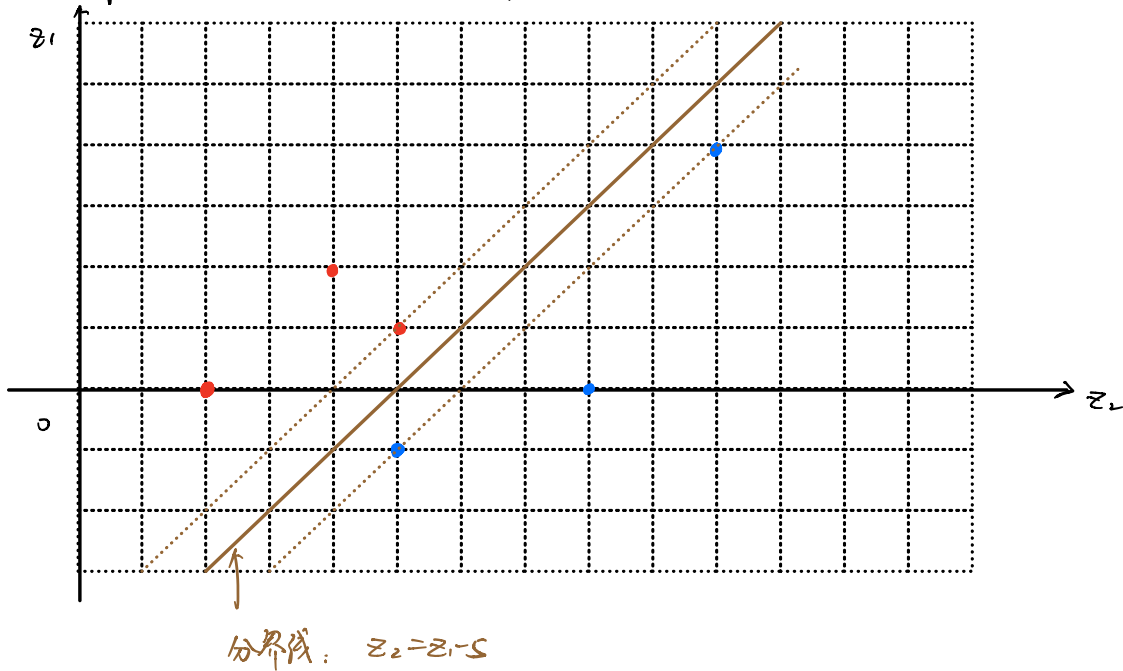
$$D_3 = (2, 0)^T \Rightarrow z = (4, 2) \quad 1$$

$$D_4 = (1, 2)^T \Rightarrow z = (5, -1) \quad -1$$

$$D_5 = (2, 2)^T \Rightarrow z = (8, 0) \quad -1$$

$$D_6 = (1, -3)^T \Rightarrow z = (10, +4) \quad -1$$

12) 如图, "•"代表+1类, "•"代表-1类,



$$13) \quad z_2 = z_1 - 5 \Rightarrow x_1 - x_2 = x_1^2 + x_2^2 - 5$$

$$\Rightarrow \left(x_1 - \frac{1}{2}\right)^2 + \left(x_2 + \frac{1}{2}\right)^2 = \frac{11}{2}$$

